

# Chapter 11: Probability

5 lessons · one lesson for each exercise · each lesson is one hour

**NSW syllabus outcome**

Stage 4 Statistics and probability: MA4-PRO-C-01 solves problems involving the probabilities of simple chance experiments.

NAME

CLASS

CHAPTER

**11**

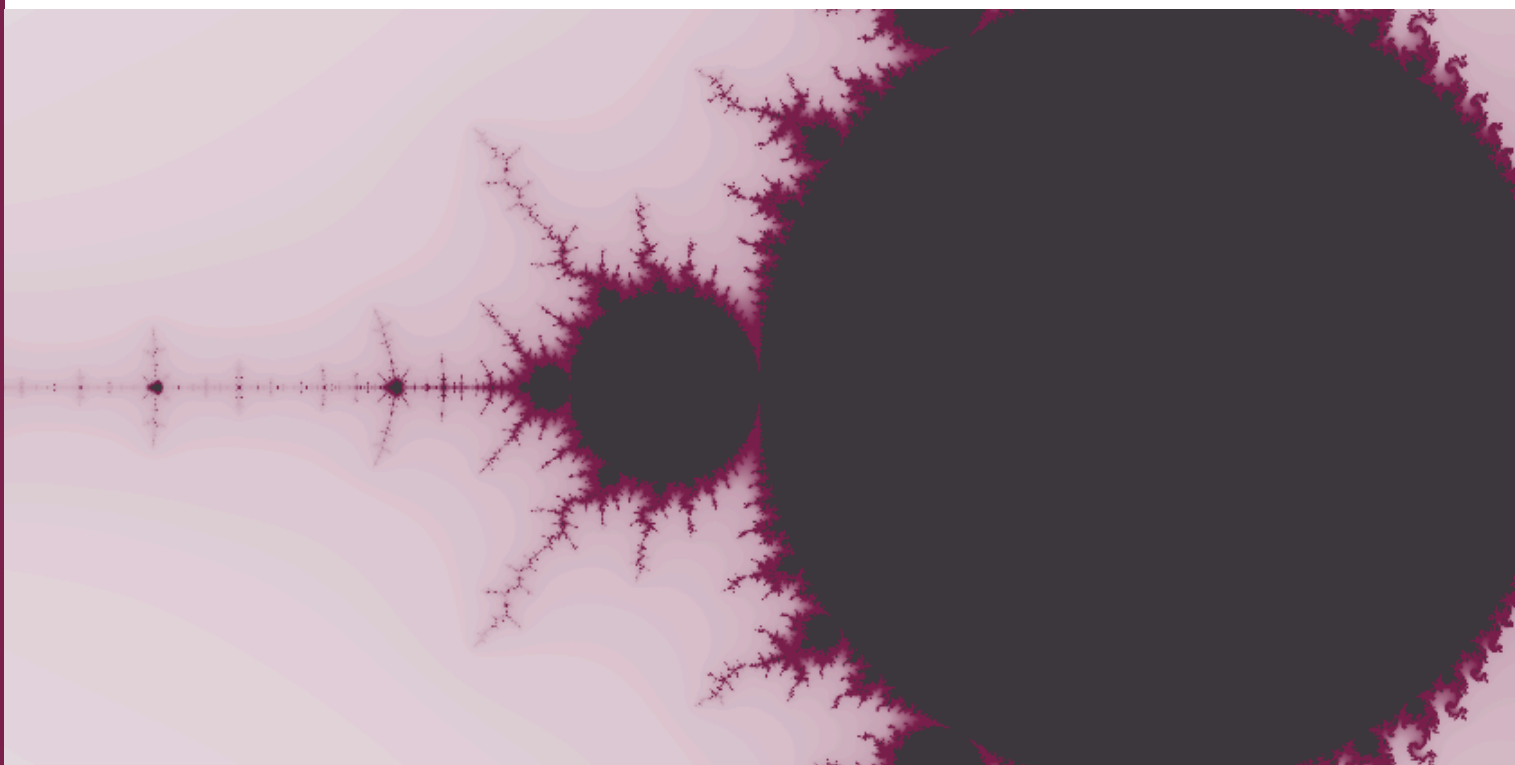
**11.01** Sample space

**11.02** Probability

**11.03** The probability scale

**11.04** Experimental probability

**11.05** Complementary events



## How to use this booklet

**1** Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

**2** Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

**WE DO** **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

**YOU DO** **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

**3** Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

**4** Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

## Contents and progress tracker

| Lesson       | Topic                    | Page      | Date | Done                     |
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| <b>11.02</b> | Probability              | <b>7</b>  |      | <input type="checkbox"/> |
| <b>11.03</b> | The probability scale    | <b>11</b> |      | <input type="checkbox"/> |
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## In this chapter you will

- list the sample space of an experiment
- find the probability of simple events as fractions, decimals and percentages
- use the probability scale and chance words
- compare experimental and theoretical probability
- find the probability of complementary events

### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule,  $z \rightarrow z^2 + c$ , thousands of times for every point. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON  
**11.01**

# Sample space

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- list the sample space of a chance experiment
- decide whether outcomes are equally likely

## Success criteria

I can:

- ☐ list every possible outcome
- ☐ write the sample space in braces { }
- ☐ explain why outcomes are or are not equally likely

**Key terms** Write the definition as your teacher explains it.

**Outcome**

**Sample space**

**Equally likely**

Notes

## Level 1 · Getting started

Easy · 17 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** List the sample space.

**a** List the sample space for tossing a coin.

Your teacher works this out. Copy every step.

**b** List the sample space for rolling a die.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: List the sample space.

12 QUESTIONS

**REMEMBER** An **OUTCOME** is one possible result.

**1** Spinning a spinner with A, B, C, D

\_\_\_\_\_

**2** Choosing a day of the week

\_\_\_\_\_

**3** Choosing a vowel

\_\_\_\_\_

**4** Rolling a die

\_\_\_\_\_

**5** The result of a soccer match for one team

\_\_\_\_\_

**6** The season of a person's birthday

\_\_\_\_\_

**Questions 7–12: how many outcomes are in the sample space?**

**7** tossing a coin

\_\_\_\_\_

**8** rolling a die

\_\_\_\_\_

**9** choosing a month

\_\_\_\_\_

**10** choosing a letter of the alphabet

\_\_\_\_\_

**11** spinning A, B, C, D, E

\_\_\_\_\_

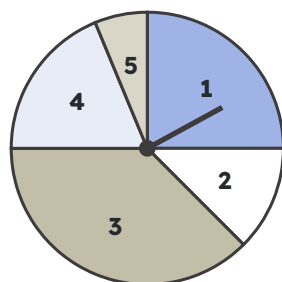
**12** choosing a card suit

\_\_\_\_\_

**YOU DO**

On your own: **Easy** Set B. Use the spinner.

5 QUESTIONS



- 1 How many outcomes?
- 2 Why aren't they equally likely?
- 3 Which number is most likely?
- 4 Which is least likely?
- 5 Which number has a probability of 25%?



**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

List all the possible outcomes.

- a** List the outcomes of a driving test.

Your teacher works this out. Copy every step.

- b** List the classifications of a new film.

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. List the sample space. Are the outcomes equally likely?

6 QUESTIONS

**REMEMBER** The **SAMPLE SPACE** is the list of all outcomes.

- 1 The public transport students use to get to school

- 3 Choosing a letter from the word APPLE

- 5 The colour of a traffic light when you arrive

- 2 Choosing a letter from the word MATHS

- 4 The number of heads when 2 coins are tossed

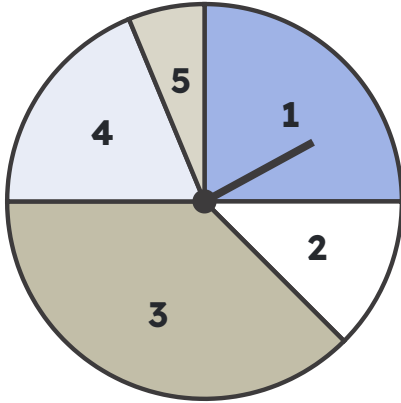
- 6 Drawing a card suit from a deck

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Use the spinner, then list the outcomes.



**a** How many outcomes? Are they equally likely?

Your teacher works this out. Copy every step.

**b** List the sample space for the winner when Katrina plays Biljana at tennis.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Two-step experiments.

4 QUESTIONS

**REMEMBER** Write it in braces: {H, T}.

**1** List the sample space for tossing 2 coins (e.g. HT).

**3** How many outcomes are there for rolling 2 dice? Explain.

**2** List all outcomes for spinning spinner ABCD and tossing a coin.

**4** Design a spinner with 4 outcomes that are not equally likely.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

List the sample space for choosing a primary colour.

**Level 2**

How many outcomes are there when rolling a die?

**Level 3**

Are the outcomes of a test (pass/fail) equally likely? Explain.

 Extension · Stretch yourself Worked example

How many outcomes are there when 3 coins are tossed? List them.

1. Each coin has 2 outcomes:  $2 \times 2 \times 2$
2. HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

Answer: 8 outcomes

## YOU DO

On your own: read the worked example, then try these.

- E1 How many outcomes when 4 coins are tossed?

- E2 How many different 2-letter codes can be made from A, B, C if letters can repeat?

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. An **outcome** is one possible result.
  2. The **sample space** is the list of all outcomes.
  3. Write it in braces: {H, T}.
  4. Outcomes are **equally likely** if each has the same chance.

 Level 1

Tossing a coin

1. 2 outcomes

Answer: {H, T}

 Level 2

Rolling a die

1. 6 outcomes

Answer: {1, 2, 3, 4, 5, 6}

 Level 3

Spinner with unequal sectors

1. Different sector sizes

Answer: not equally likely

 Extension

3 coins

1.  $2 \times 2 \times 2$

Answer: 8 outcomes

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LESSON  
11.02

# Probability

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the probability of a simple event

## Success criteria

I can:

- ☐ use  $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total outcomes}}$
- ☐ write probability as a fraction, decimal or percentage
- ☐ round percentages to 1 decimal place

**Key terms** Write the definition as your teacher explains it.

**Probability**

**Event**

**At random**

Notes

## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** A die is rolled. Find each probability.

**a** On a die, find  $P(1)$ .

**b** On a die, find  $P(\text{odd number})$ .

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: A die is rolled. Find the probability of rolling: **12 QUESTIONS**

**REMEMBER**  $P(\text{event}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ .

**1** a 6

**2** an even number

**3** a number less than 5

**4** a 7

**5** a number greater than 0

**6** a 2 or a 3

Questions 7–12: a bag has 3 red, 5 blue and 2 green counters. Find:

**7**  $P(\text{red})$

**8**  $P(\text{blue})$

**9**  $P(\text{green})$

**10**  $P(\text{red or green})$

**11**  $P(\text{yellow})$

**12**  $P(\text{not blue})$

**YOU DO**

On your own: **Easy** Set B. A letter is chosen at random from PROBABILITY. Find the probability of choosing:

6 QUESTIONS

1 B

4 Y

2 I

5 a letter in the word ROB

3 a vowel

6 Z



**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Find each probability.

**a** On a die, find  $P(\text{number less than 5})$ .

Your teacher works this out. Copy every step.

**b** A box has 15 dark, 13 milk and 12 white chocolates. Find  $P(\text{white})$ .

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. Find each probability as a fraction and a percentage (1 d.p.).

6 QUESTIONS

**REMEMBER** Probabilities are between 0 and 1.

1 15 dark, 13 milk, 12 white chocolates:  $P(\text{white})$

3  $P(\text{not milk})$

5  $P(\text{exercise books or paper})$

2  $P(\text{dark})$

4 Truck (1040 boxes):  $P(\text{rulers})$

6  $P(\text{erasers})$

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find each probability as a percentage (1 d.p.).

- a** A truck has 325 boxes of books, 210 of rulers, 360 of paper and 145 of pens. Find  $P(\text{rulers})$  as a % (1 d.p.).

Your teacher works this out. Copy every step.

- b** Using the truck, find  $P(\text{no pens})$  as a % (1 d.p.).

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER** Simplify fractions.

- 1** A class has 14 girls and 11 boys. One student is chosen at random. Find  $P(\text{boy})$  as a decimal.

- 3** A bag has red and blue counters.  $P(\text{red}) = 0.35$ . What is  $P(\text{blue})$ ?

- 2** A raffle sells 250 tickets. You buy 4. Find your probability of winning as a percentage.

- 4** A bag has 12 counters and  $P(\text{green}) = \frac{1}{4}$ . How many are green?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find  $P(5)$  on a die.

**Level 2**

A bag has 3 red and 7 blue marbles. Find  $P(\text{red})$ .

**Level 3**

Write your answer to Question 2 as a percentage.

**Extension · Stretch yourself****Worked example**

A card is drawn from a standard deck of 52.  
Find  $P(\text{red king})$ .

1. There are 2 red kings (hearts, diamonds)
2.  $\frac{2}{52}$  simplifies

Answer:  $\frac{1}{26}$

**YOU DO**

On your own: read the worked example,  
then try these.

- E1** Find  $P(\text{a heart or a king})$  from a standard deck.

- E2** Find  $P(\text{a picture card: J, Q, K})$  from a standard deck.

**Lesson summary**

Missed the lesson? Start here.

**Remember**

1.  $P(\text{event}) = \text{favourable outcomes} \div \text{total outcomes}$ .
3. Simplify fractions.

2. Probabilities are between 0 and 1.

4. Percentage:  $\times 100\%$ , round as asked.

**Level 1**

$P(\text{odd})$  on a die

1. 3 odd of 6

Answer:  $\frac{1}{2}$

**Level 2**

$P(\text{white chocolate})$

1. 12 of 40

Answer:  $\frac{3}{10}$  (C)

**Level 3**

$P(\text{rulers})$

1.  $210 \div 1040$

Answer: 20.2%

**Extension**

$P(\text{red king})$

1. 2 of 52

Answer:  $\frac{1}{26}$

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LESSON  
11.03

## The probability scale

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- describe chance using words and the probability scale

## Success criteria

I can:

- ☐ place events on a scale from 0 (impossible) to 1 (certain)
- ☐ match probabilities to words such as unlikely and good chance
- ☐ find probabilities from a jar of objects

**Key terms** Write the definition as your teacher explains it.**Certain****Impossible****Even chance**

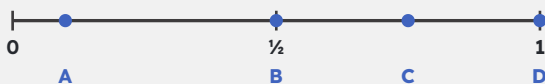
Notes

## Level 1 · Getting started

Easy · 12 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Use the probability scale and chance words.

- a** Where on the scale is the chance that the next baby is a girl?



Your teacher works this out. Copy every step.

- b** Which word describes a probability of 0.29?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Describe each event using a chance word.

6 QUESTIONS

**REMEMBER** Probability goes from 0 (impossible) to 1 (certain).

- |                                               |                                          |                                               |
|-----------------------------------------------|------------------------------------------|-----------------------------------------------|
| 1 The sun will rise tomorrow.<br>_____        | 2 Rolling a 7 on a die.<br>_____         | 3 A coin lands on heads.<br>_____             |
| 4 It will snow in Sydney in January.<br>_____ | 5 Rolling a number less than 6.<br>_____ | 6 Your teacher is over 10 years old.<br>_____ |

**YOU DO**

On your own: **Easy** Set B. Match each probability to a word: impossible, unlikely, even chance, likely, certain. **6 QUESTIONS**

1 0

4 0.85

2 0.29

5 1

3  $\frac{1}{2}$

6 15%

**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Find each probability.

- a** Which is certain: choosing a 4 from {1, 2, 3, 4}; a red ball from green balls; an odd or even number on a die; a baby being a girl?

Your teacher works this out. Copy every step.

- b** A jar has 1 red, 6 yellow, 2 white and 5 black jellybeans. Find  $P(\text{yellow})$ .

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. A jar has 1 red, 6 yellow, 2 white and 5 black jellybeans. Find:

**6 QUESTIONS**

**REMEMBER**  $\frac{1}{2}$  is an even chance.

1  $P(\text{yellow})$

3  $P(\text{blue})$

5  $P(\text{black})$

2  $P(\text{white or red})$

4  $P(\text{not green})$

6  $P(\text{not yellow})$

**Level 3 · Challenge**

Challenging · 2 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find each probability.

**a** Find  $P(\text{white or red})$ .

Your teacher works this out. Copy every step.

**b** Find  $P(\text{blue})$  and  $P(\text{not green})$ .

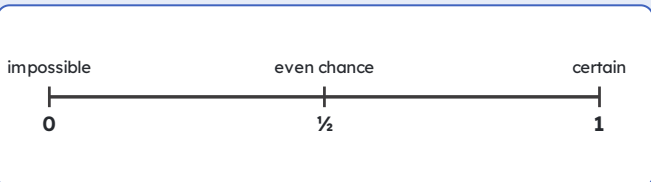
Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Mark each event on the scale.

2 QUESTIONS

**REMEMBER** Less than  $\frac{1}{2}$ : unlikely. More than  $\frac{1}{2}$ : likely.

**1** Mark: A rolling an even number; B rolling a 6; C rolling a number from 1 to 6; D rolling a 0.



**2** Describe an event that is: (a) impossible (b) unlikely (c) certain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

What is the probability of an impossible event?

**Level 2**

Which word describes  $P = 0.9$ ?

**Level 3**

A bag has 3 red and 5 blue. Find  $P(\text{not red})$ .

**Extension · Stretch yourself**

**Worked example**

A bag has 20 counters.  $P(\text{red}) = 0.4$  and  $P(\text{blue}) = 0.35$ . The rest are green. How many are green?

1.  $P(\text{green}) = 1 - 0.4 - 0.35 = 0.25$

2.  $0.25 \times 20$

**Answer:** 5

**YOU DO**

On your own: read the worked example, then try these.

- E1** A spinner has  $P(\text{red}) = \frac{1}{3}$ . It has 12 equal sectors. How many are red?

- E2** Design a bag of 10 marbles where  $P(\text{blue})$  is “unlikely” and  $P(\text{red})$  is “likely”.

**Lesson summary**

Missed the lesson? Start here.

**Remember**

1. Probability goes from 0 (impossible) to 1 (certain).
3. Less than  $\frac{1}{2}$ : unlikely. More than  $\frac{1}{2}$ : likely.

2.  $\frac{1}{2}$  is an even chance.

4. Count the objects to find each probability.

**Level 1**

Baby is a girl

1. About half

**Answer:** B (even chance)

**Level 2**

0.29

1. Less than  $\frac{1}{2}$

**Answer:** unlikely

**Level 3**

$P(\text{yellow})$

1. 6 of 14

**Answer:**  $\frac{3}{7}$

**Extension**

Green counters

1.  $0.25 \times 20$

**Answer:** 5

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
**11.04**

# Experimental probability

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- compare experimental and theoretical probability

## Success criteria

I can:

- ☐ find theoretical probability from equally likely outcomes
- ☐ find experimental (observed) probability from results
- ☐ find expected frequency = probability × number of trials

**Key terms** Write the definition as your teacher explains it.

**Experimental probability**

**Expected frequency**

**Trial**

Notes

## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the theoretical probability and the expected number.

- a** A coin is tossed 80 times. What is the theoretical  $P(\text{head})$ ?

- b** What is the expected number of heads in 80 tosses?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Find the theoretical probability.

12 QUESTIONS

**REMEMBER** Theoretical  $P = \text{favourable} \div \text{total}$  (equally likely outcomes).

- 1**  $P(\text{head})$  on a coin

- 2**  $P(4)$  on a die

- 3**  $P(A)$  on spinner ABCD

- 4**  $P(\text{even})$  on a die

- 5**  $P(\text{red})$  from 3 red, 2 blue

- 6**  $P(\text{vowel})$  from A, B, C, D, E

Questions 7–12: find the expected number.

- 7** heads in 50 tosses

- 8** 6s in 30 rolls

- 9** even numbers in 40 rolls

- 10** A on spinner ABCD in 100 spins

- 11** tails in 200 tosses

- 12** a 1 or 2 in 60 rolls

**YOU DO** On your own: **Easy** Set B. Find the expected frequency.

6 QUESTIONS

- |                            |                      |                                              |                      |
|----------------------------|----------------------|----------------------------------------------|----------------------|
| 1 Heads in 80 tosses       | <input type="text"/> | 4 A on spinner ABCD in 200 spins             | <input type="text"/> |
| 2 6s in 60 rolls           | <input type="text"/> | 5 Red in 100 draws (3 red, 2 blue, replaced) | <input type="text"/> |
| 3 Even numbers in 50 rolls | <input type="text"/> | 6 Tails in 1000 tosses                       | <input type="text"/> |

**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Find the observed probability, then compare.

- a** It landed heads 24 times. What is the observed probability of a head?
- b** How does the observed frequency compare with the expected frequency?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Find the observed (experimental) probability.

6 QUESTIONS

**REMEMBER** Observed P = number of times it happened ÷ number of trials.

- |                         |                        |                        |
|-------------------------|------------------------|------------------------|
| 1 24 heads in 80 tosses | 3 42 reds in 100 draws | 5 130 A's in 400 spins |
| <input type="text"/>    | <input type="text"/>   | <input type="text"/>   |
| 2 15 sixes in 60 rolls  | 4 7 wins in 20 games   | 6 0 fives in 12 rolls  |
| <input type="text"/>    | <input type="text"/>   | <input type="text"/>   |

**Level 3 · Challenge**

Challenging · 2 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the expected number, then explain.

**a** A die is rolled 60 times. How many 6s are expected?

Your teacher works this out. Copy every step.

**b** Why do more trials give a better estimate?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Do an experiment. Toss a coin 20 times (or use results from your class). **2 QUESTIONS**

**REMEMBER** Expected frequency =  $P \times \text{number of trials}$ .

**1** Record your results in a tally table.

| Outcome | Tally | Frequency |
|---------|-------|-----------|
| Heads   |       |           |
| Tails   |       |           |

**2** Find your observed  $P(\text{head})$ . Compare it with the theoretical probability. Combine with the class results. What do you notice?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

What is the theoretical  $P(3)$  on a die?

**Level 2**

How many 3s are expected in 120 rolls?

**Level 3**

A 3 came up 25 times in 120 rolls. Find the observed  $P(3)$ .

**Extension · Stretch yourself****Worked example**

A spinner landed on red 45 times in 150 spins. Estimate how many times it will land on red in 1000 spins.

1. Observed  $P(\text{red}) = 45 \div 150 = 0.3$
2.  $0.3 \times 1000$

**Answer:** about 300 times

**YOU DO**

On your own: read the worked example, then try these.

- E1** A drawing pin landed point up 36 times in 60 throws. Estimate how many times it lands point up in 250 throws.

- E2** Is this spinner likely to have equal sectors if there are 4 colours? Explain.

**Lesson summary**

Missed the lesson? Start here.

- Remember**
1. Theoretical  $P = \text{favourable} \div \text{total}$  (equally likely outcomes).
  3. Expected frequency =  $P \times \text{number of trials}$ .

2. Observed  $P = \text{number of times it happened} \div \text{number of trials}$ .
4. More trials: observed  $P$  gets closer to theoretical  $P$ .

**Level 1**

Theoretical  $P(\text{head})$

1. 1 of 2

**Answer:**  $\frac{1}{2}$

**Level 2**

Expected heads in 80

1.  $\frac{1}{2} \times 80$

**Answer:** 40

**Level 3**

Observed  $P(\text{head})$

1.  $24 \div 80$

**Answer:**  $\frac{3}{10}$

**Extension**

Red in 1000 spins

1.  $0.3 \times 1000$

**Answer:** about 300

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LESSON  
11.05

## Complementary events

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the probability of complementary events

## Success criteria

I can:

- ☐ describe the **complement** of an event (“not” the event)
- ☐ use  $P(\text{not } E) = 1 - P(E)$
- ☐ solve problems using complementary events

**Key terms** Write the definition as your teacher explains it.**Complementary events****Complement**

Notes

## Level 1 • Getting started

Easy • 18 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Write the complement and its probability.

- a** Write the complement of selecting a queen from a deck, and its probability.

Your teacher works this out. Copy every step.

- b** Write the complement of rolling a multiple of 3 on a die, and its probability.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Write the complementary event.

12 QUESTIONS

**REMEMBER** The complement of E is “not E”.

- 1** Rolling a 6

\_\_\_\_\_

- 2** Getting a head

\_\_\_\_\_

- 3** Drawing a red card

\_\_\_\_\_

- 4** Winning a game (no draws)

\_\_\_\_\_

- 5** Choosing a vowel

\_\_\_\_\_

- 6** Rolling an even number

\_\_\_\_\_

Questions 7–12: find  $P(\text{not } E)$ .

- 7**  $P(E) = 0.3$

\_\_\_\_\_

- 8**  $P(E) = \frac{1}{5}$

\_\_\_\_\_

- 9**  $P(E) = 45\%$

\_\_\_\_\_

- 10**  $P(E) = 0$

\_\_\_\_\_

- 11**  $P(E) = \frac{3}{8}$

\_\_\_\_\_

- 12**  $P(E) = 0.99$

\_\_\_\_\_

**YOU DO**

On your own: **Easy** Set B. Find  $P(\text{not } E)$ .

## 6 QUESTIONS

①  $P(E) = \frac{1}{4}$

④  $P(E) = \frac{5}{12}$

2  $P(E) = 0.3$

5  $P(E) = 0.07$

3  $P(E) = 45\%$

6  $P(E) = 1$

\_\_\_\_\_

 **Level 2 · Building up**

Medium · 6 questions

**WE DO**

**With your teacher: copy what they write on the board.**

### Example 2

**Write the complement and its probability, then find P.**

- a** Write the complement of buying the winning ticket out of 550 sold, and its probability.

Your teacher works this out. Copy every step.

- b** Names Ashleigh, Hamish, Sarah, Jake, Hayley: find  $P(\text{Hamish})$ .

Your teacher works this out. Copy every step.

**YOU DO**

**On your own: Medium** Set C. Find each probability.

## 6 QUESTIONS

**REMEMBER** E and not E cover all outcomes.

- ① P(not a queen) from a deck

[illegible]

- ③  $P(\text{not winning})$  with 1 ticket of 550

[illegible]

- 5 P(not brown): 8 brown, 4 black, 3 white

- ② P(not a multiple of 3) on a die

- ④ P(not Hamish) from the 5 names

[illegible]

- 6  $P(\text{not a heart})$  from a deck



## Level 3 · Challenge

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.**Example 3** Find each probability.

- a** Find  $P(\text{a girl's name})$  and  $P(\text{not beginning with H})$ .

Your teacher works this out. Copy every step.

- b** A bag has 8 brown, 4 black and 3 white marbles. Find  $P(\text{not brown})$ .

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Problems.

4 QUESTIONS

**REMEMBER**  $P(\text{not } E) = 1 - P(E)$ .

- 1** The probability it will rain tomorrow is 0.35. What is the probability it will not rain?

- 3** In a class,  $P(\text{student has a pet}) = 17/25$ . There are 25 students. How many do not have a pet?

- 2**  $P(\text{a team wins}) = 0.6$  and  $P(\text{draw}) = 0.1$ . Find  $P(\text{the team loses})$ .

- 4** Explain why  $P(E) + P(\text{not } E) = 1$ .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Write the complement of rolling a 2.

**Level 2**If  $P(E) = 0.8$ , find  $P(\text{not } E)$ .**Level 3**A bag has 5 red, 3 blue and 2 green. Find  $P(\text{not red})$ .

**Extension · Stretch yourself****Worked example**

A spinner has  $P(\text{red}) = 0.2$ ,  $P(\text{blue}) = 0.45$ ,  $P(\text{green}) = \text{the rest}$ . Find  $P(\text{not blue})$  and  $P(\text{green})$ .

1.  $P(\text{not blue}) = 1 - 0.45$
2.  $P(\text{green}) = 1 - 0.2 - 0.45$

**Answer:**  $P(\text{not blue}) = 0.55$ ,  $P(\text{green}) = 0.35$

**YOU DO**

On your own: read the worked example, then try these.

- E1** A bag has red, white and blue marbles.  $P(\text{red}) = \frac{1}{3}$  and  $P(\text{white}) = \frac{1}{4}$ . Find  $P(\text{blue})$ .

- E2** In 200 spins of the spinner above, how many times would you expect NOT green?

**Lesson summary**

Missed the lesson? Start here.

**Remember**

1. The complement of E is “not E”.
2. E and not E cover all outcomes.
3.  $P(\text{not E}) = 1 - P(E)$ .
4. Use it when “not” is easier to count.

**Level 1**

Not a queen

1. 4 queens of 52

**Answer:**  $\frac{12}{13}$

**Level 2**

Not a multiple of 3

1.  $1 - \frac{1}{3}$

**Answer:**  $\frac{2}{3}$

**Level 3**

Not brown

1. 7 of 15

**Answer:**  $\frac{7}{15}$

**Extension**

Spinner

1.  $1 - 0.45$

**Answer:** 0.55

This space is left blank on purpose. It is the back of your exit ticket.